

Goal Classifier

Felipe Rotelli Prado Moreira



Dataset

- Used StatsBomb open dataset
- Free Version
- Access to 75 different seasons from many different competitions

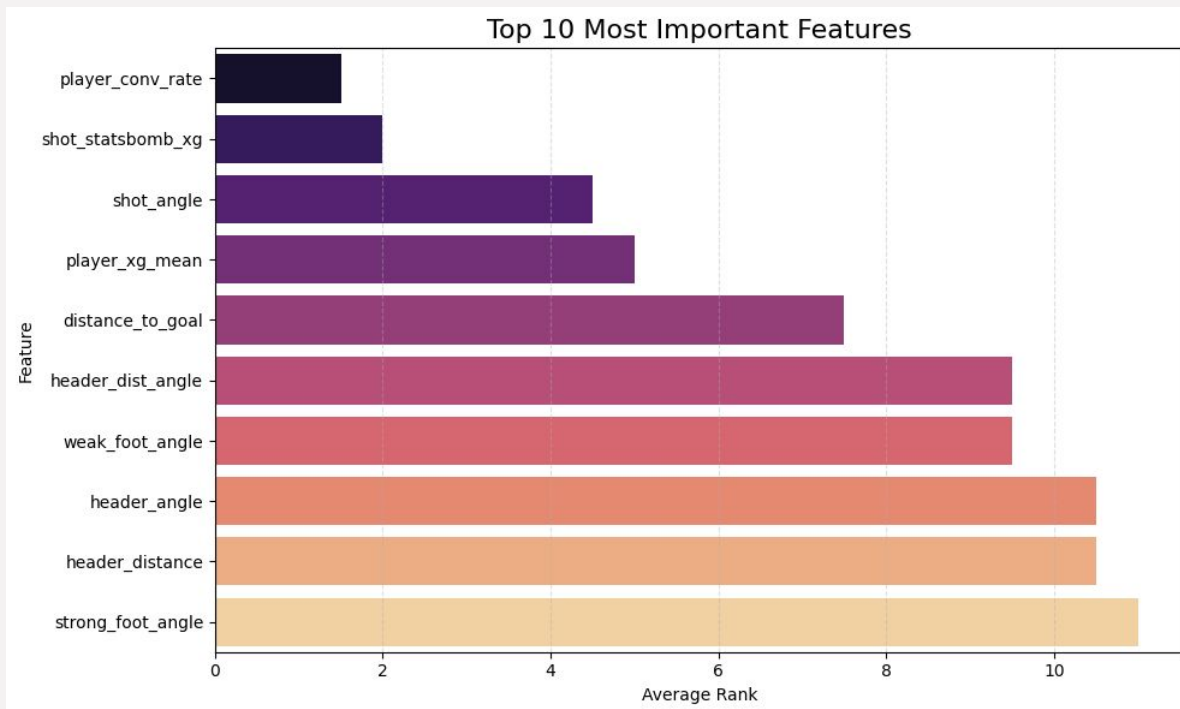


Top Five Leagues



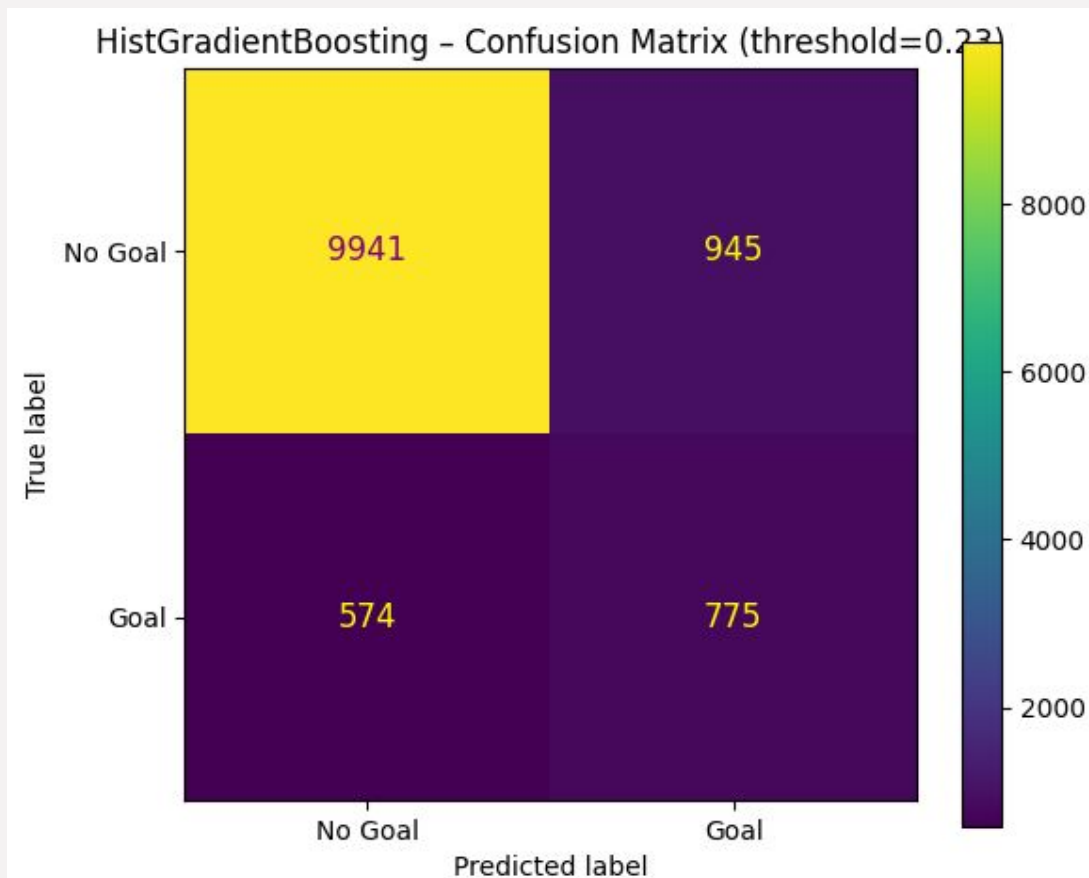
Features Used

- Used 21 different features
- Most Important feature:
 - Player conversion rate
- Accounted for Headers Differently



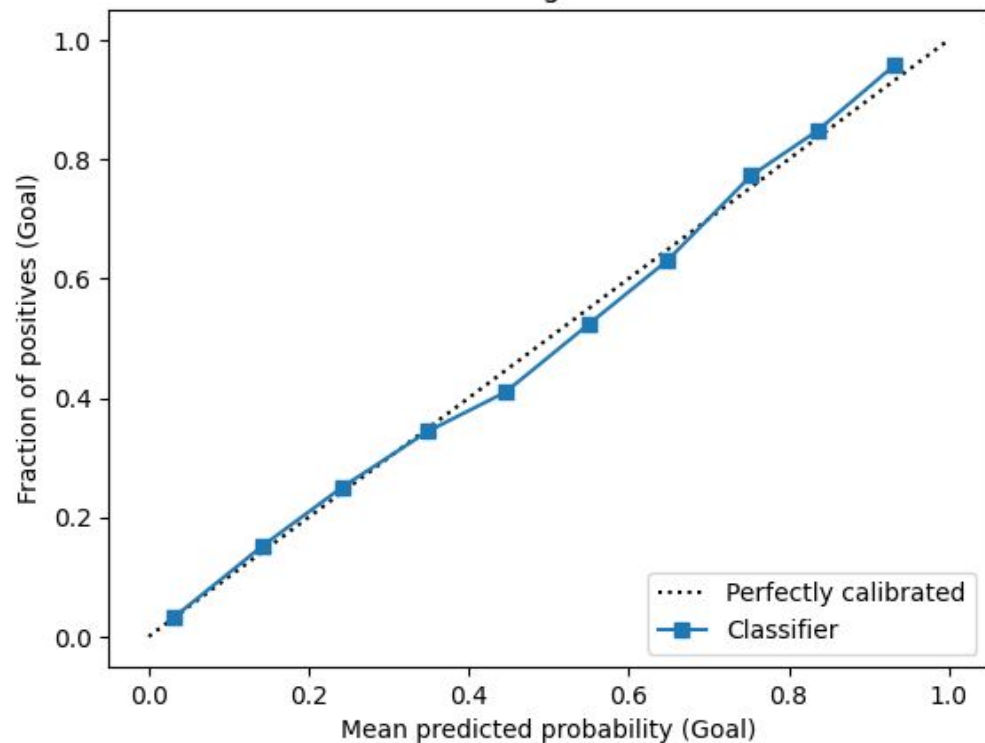
Results

- Model predicted correctly when it wasn't a goal 9,941 time out of 10,886 no goals
- 91%
- For Goals predicted 775 out of 1,349 goals
- 57%

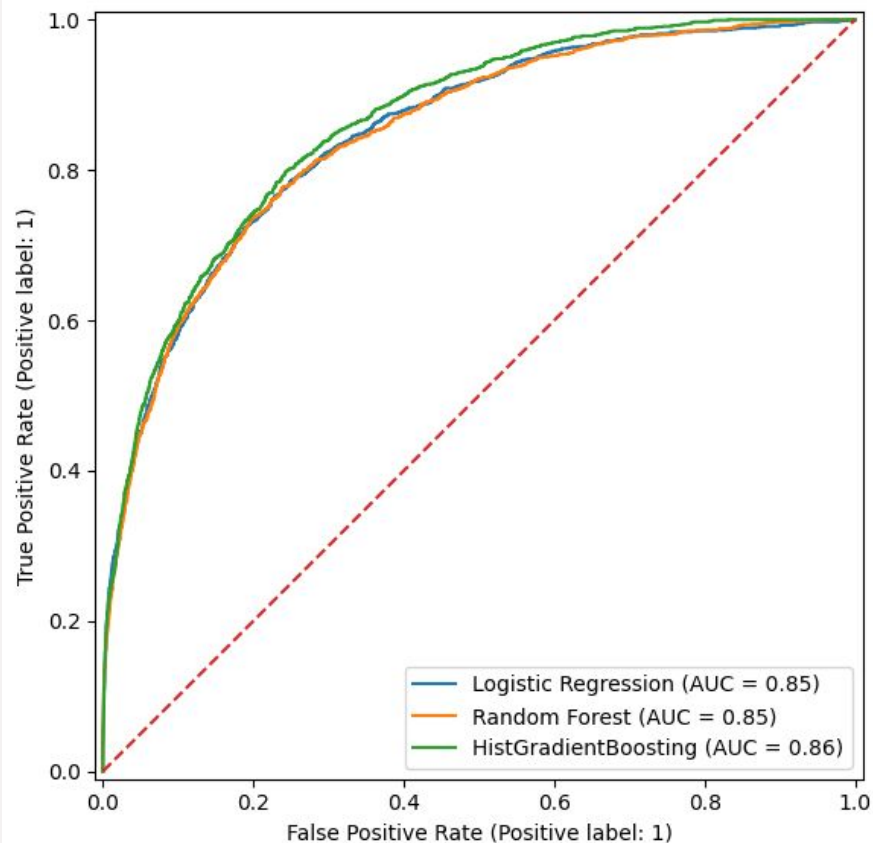


Results

HistGradientBoosting - Calibration Curve

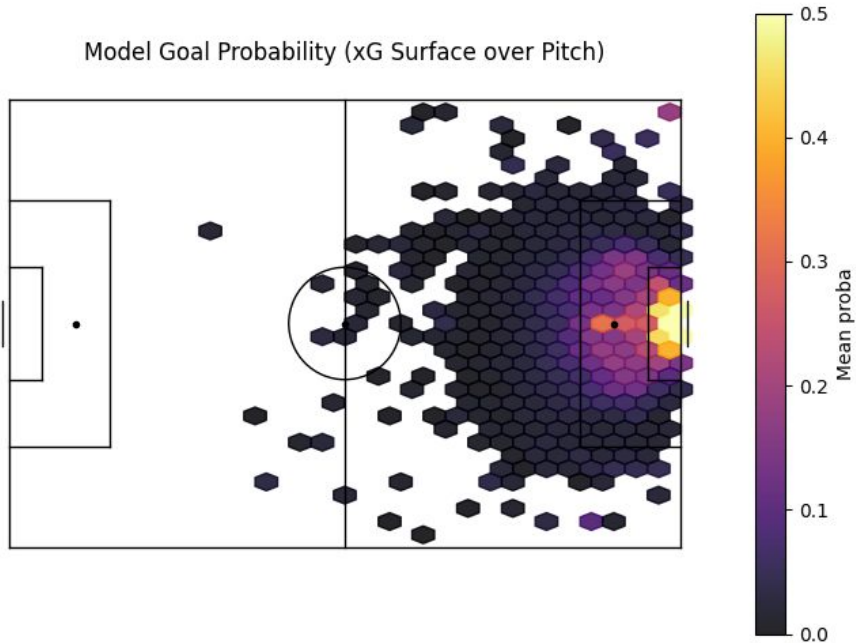


ROC Curves — Goal Classifier

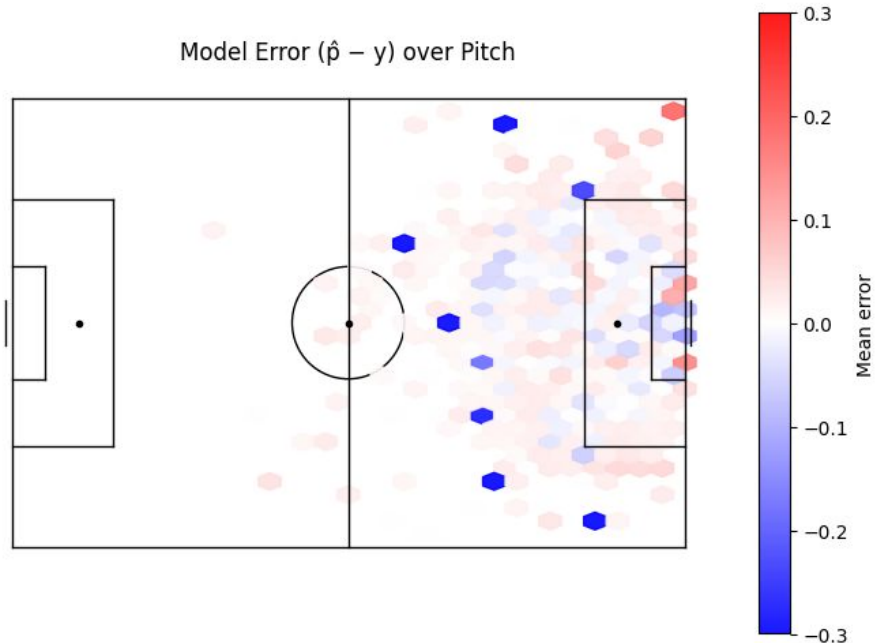


Results

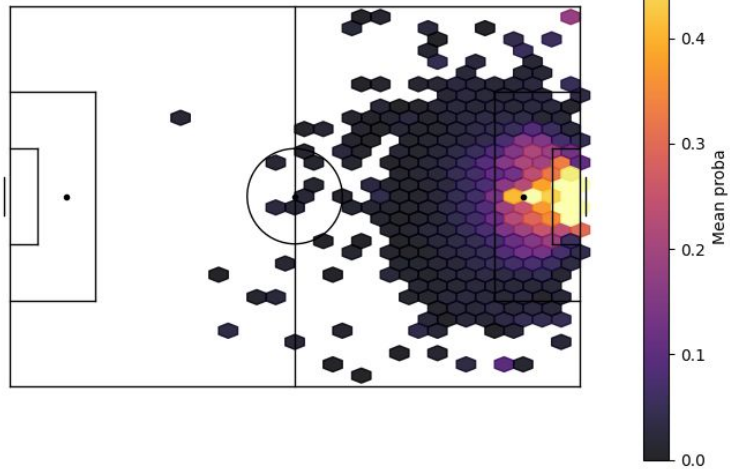
Model Goal Probability (xG Surface over Pitch)



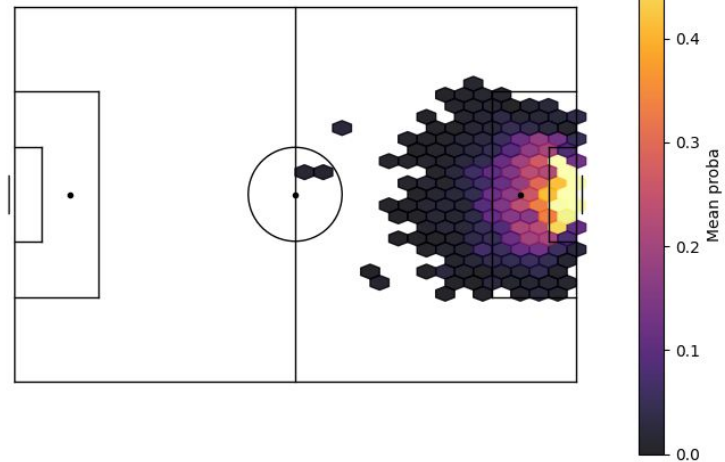
Model Error ($\hat{p} - y$) over Pitch



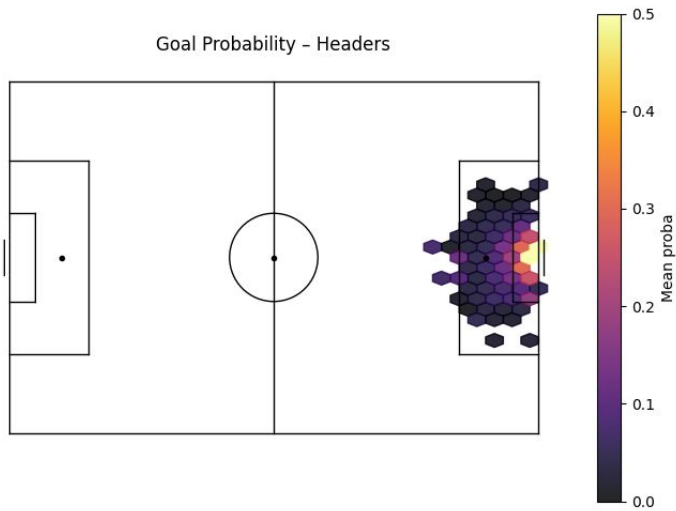
Goal Probability - Strong Foot (Non-headers)



Goal Probability - Weak Foot (Non-headers)



Goal Probability - Headers



Players Model Got Wrong

Model Overestimated

Top players model overestimates (believes in too much):

	player_name	n_shots	goals	avg_pred_proba	empirical_conv_rate	model_minus_empirical
1298	Ronny Rodelin	17	0	0.160321	0.000000	0.160321
1501	Massimo Maccarone	18	0	0.158259	0.000000	0.158259
117	Alexandre Lacazette	23	2	0.237783	0.086957	0.150827
333	Carlos Arturo Bacca Ahumada	20	2	0.223744	0.100000	0.123744
355	Charles Días Barbosa de Oliveira	17	1	0.168525	0.058824	0.109702
1804	Pierre-Emerick Aubameyang	25	3	0.229361	0.120000	0.109361
1537	Mauro Emanuel Icardi Rivero	16	2	0.230732	0.125000	0.105732
1241	Kevin Gameiro	17	3	0.281598	0.176471	0.105127
1960	Sadio Mané	18	2	0.211429	0.111111	0.100318
1539	Mauro Matías Zárate	17	0	0.099333	0.000000	0.099333
656	Federico Dionisi	18	0	0.090556	0.000000	0.090556
1104	Jordi Alba Ramos	24	0	0.089392	0.000000	0.089392
1252	Kevin Volland	19	2	0.193949	0.105263	0.088685
1432	Marco Borriello	17	1	0.140114	0.058824	0.081291
2031	Shane Long	15	2	0.214329	0.133333	0.080996

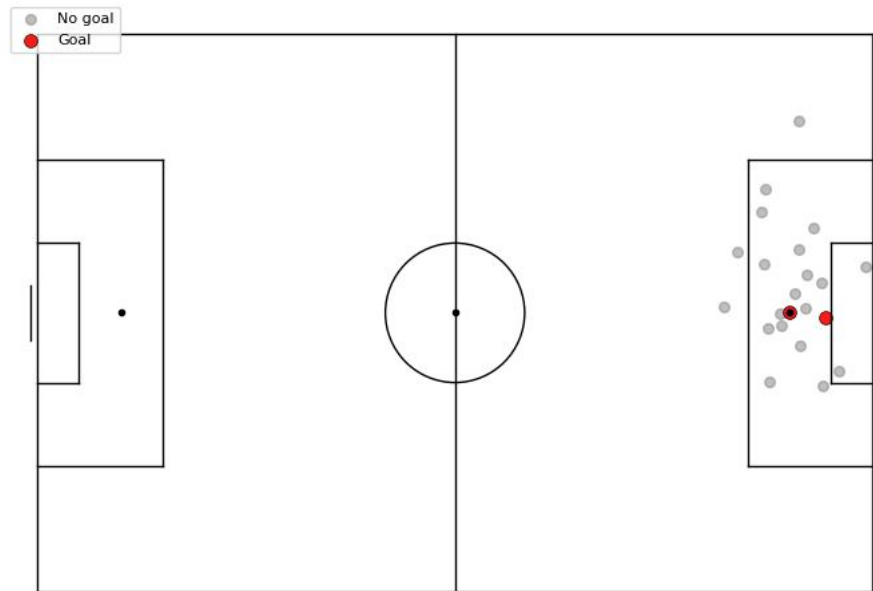
Model Underestimated

Top players model underestimates (doubts too much):

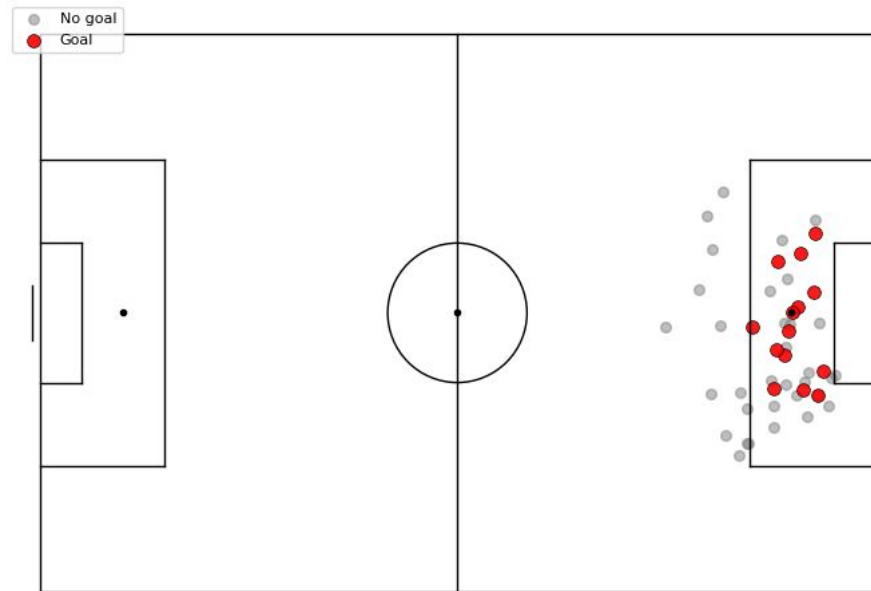
	player_name	n_shots	goals	avg_pred_proba	empirical_conv_rate	model_minus_empirical
168	André Ayew Pelé	18	5	0.128381	0.277778	-0.149397
674	Fernando Llorente Torres	17	4	0.090041	0.235294	-0.145253
365	Cheikh N"Doye	18	5	0.143043	0.277778	-0.134734
839	Graziano Pellè	21	6	0.164820	0.285714	-0.120894
1272	Kylian Mbappé Lottin	47	14	0.180063	0.297872	-0.117810
1829	Raffael Caetano de Araújo	19	4	0.099936	0.210526	-0.110590
1090	Jonathan Viera Ramos	18	5	0.169773	0.277778	-0.108005
1573	Michy Batshuayi Tunga	26	6	0.127045	0.230769	-0.103725
1886	Robert Pirès	16	7	0.339017	0.437500	-0.098483
318	Caiuby Francisco da Silva	21	3	0.045018	0.142857	-0.097839
815	Giuseppe Rossi	16	3	0.090829	0.187500	-0.096671
213	Aritz Aduriz Zubeldia	28	6	0.120972	0.214286	-0.093313
590	Emiliano Sala	19	4	0.119622	0.210526	-0.090904
1308	Leonardo Pavoletti	18	5	0.188873	0.277778	-0.088905
232	Arturo Erasmo Vidal Pardo	23	5	0.135118	0.217391	-0.082274

Model Overestimates vs Underestimates – Shot Maps

Overestimated: Alexandre Lacazette
Model p: 0.24 | Empirical: 0.09 | $\Delta=0.15$



Underestimated: Kylian Mbappé Lottin
Model p: 0.18 | Empirical: 0.30 | $\Delta=-0.12$



**Thank
You!**